

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP40020
Subject Title	Integrated Capstone Project
Credit Value	6
Level	4
Pre-requisite / Co-requisite/ Exclusion	Nil.
Objectives	The objectives are to exert the student's ability in: - organizing a complete scientific, engineering and educational task that integrates the knowledge and skills in artificial intelligence and data analytics (AIDA), applied physics, materials science/technology, electrical or electronic engineering; - conducting systematic studies on physical mechanisms, theoretical analyses, numerical computations, device developments and/or system integrations using AIDA, applied physics, materials, electrics and/or electronics; - critical thinking and problem-solving skills in AIDA, physics, materials and engineering; - communication skills and team working spirits for various professions; - presenting the project results and writing project reports.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> - plan and successfully complete a project relevant to the selected problem area in AIDA, physics, materials and/or engineering; - conduct literature surveys to locate materials and sources, and understand the state of the art and connect the knowledge with the problem to be solved using AIDA knowledge, physics mechanisms, material science and engineering experimental skills; - assimilate and apply the learnt knowledge to generate feasible solutions to the problem; - think critically the formulation of alternative models and solutions to the problem, in the analysis of approaches to the solution and their implementation; and - evaluate the final outcome in an objective manner. - recognize the limitations of the project and make suggestions for further work; - developing broad-based problem solving skills for various professions. <u>Attributes for all-roundedness</u> - improve presentation and communication skills via oral presentations; - enhance technical report writing skills with proper organization of materials; - develop the ability to learn independently and to find/integrate information from different sources required in solving real-life problems; - manage the project efficiently and effectively through the supervision of supervisor(s); and

	(l) work collaboratively with related parties (e.g. vendors, sponsor company, technical support staff, team-partners, research students, etc.).																																																																																		
(m)Subject Synopsis/ Indicative Syllabus	1. In-depth study of a topic typically proposed by the supervisor 2. Project meeting and planning 3. Proposal writing 4. Regular progress checking and reporting 5. Project documentation 6. Presentation and demonstration Students are expected to identify a project topic with a supervisor in physics, materials and/or engineering, and a co-supervisor with AIDA expertise. Students need to demonstrate their knowledge in both AIDA and their chosen discipline in the project, receiving advice from both supervisors. The project should represent substantial efforts in analyzing and interpreting the data/information obtained, using the principles and techniques learnt from various related subjects. Students are also expected to demonstrate significant analytical and, preferably, research ability in physics, materials, engineering and applications.																																																																																		
Teaching/Learning Methodology	The Integrated Capstone Project spans across the academic year for two consecutive semesters. The teaching/learning activities include regular project meetings with the supervisor and the co-supervisor, guided study of project materials, independent project development work and other project management tasks.																																																																																		
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="12">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th><th>g</th><th>h</th><th>i</th><th>j</th><th>k</th><th>l</th></tr><tr><td>Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Project report</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Project presentation</td><td>30</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100</td><td colspan="12"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The Integrated Capstone Project will be accessed by the supervisor/co-supervisor and other assessors. Attributes to be assessed include, but not limited to, Problem Identification, Problem Solving, Scientific Merits and Engineering Significance of Project Results, Communication and Presentation, Project Management, and Attitude. Integrated Capstone Projects should be problem-oriented and there are no restrictions on the nature of the problem except that it should be relevant to AIDA, physics, materials and/or engineering. The project could be practical, academic or a hybrid in which the student is encouraged but not constrained to have some original contributions. Each student has to pass a project-starting interview, two progress interviews and a final	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)												a	b	c	d	e	f	g	h	i	j	k	l	Continuous assessment	40	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Project report	30	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Project presentation	30	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Total	100												
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	presentation, and to submit a final project report. The proposal must be approved by the supervisor/co-supervisor before the student can proceed with the Integrated Capstone Project. An oral presentation and demonstration is essential at the end of the project. Two progress interviews are required for proper continuous assessment.	
Student Study Effort Expected	Class contact:	
	Lecture	0 h
	Other student study effort:	
	Searching and reading materials, meeting with supervisor / co-supervisor / others, design and system development, testing, documentation, presentation, etc.	210 h
	Total student study effort	210 h
Reading List and References	Reference Books: L. Igual, S. Segu, “Introduction to Data Science. A Python Approach to Concepts, Techniques and Applications”, Springer (2017). References: - Kumar, Ranjit, Research Methodology: A Step-by-step Guide for Beginners, 3rd Edition, SAGE Publications, 2011. - Burns, Robert B., Introduction to Research Methods, 4th Edition, SAGE Publications, 2000. - Roberts, Carol M., The Dissertation Journey: A Practical and Comprehensive Guide to Planning, Writing, and Defending Your Dissertation, 3rd Edition, Corwin Press, 2007. - Mauch, James E. and Park, Namgi, Guide to the Successful Thesis and Dissertation: A Handbook for Students and Faculty, 5th Edition, Marcel Dekker, 2003. - Rudestam, Kjell Erik and Newton, Rae R., Surviving Your Dissertation: A Comprehensive Guide to Content and Process, 2nd Edition, Sage Publications, 2001. - Garson, G. David, Guide to Writing Empirical Papers, Theses and Dissertations, Marcel Dekker, 2002. - Reinhart, Susan M., Giving Academic Presentations, 2nd Edition, University of Michigan Press, 2013.	

	8. Oshima, Alice and Hogue, Ann, Writing Academic English, 4th Edition, Pearson Longman, 2006. 9. American Psychological Association. Publication Manual of the American Psychological Association, 6th Edition, American Psychological Association, 2010. 10. Szuchman, Lenore T., Writing with Style: APA Style Made Easy, 5th Edition, Wadsworth/Cengage Learning, 2011. 11. Statistics, simulation, programming, and relevant books. 12. ACM and IEEE magazines, Transactions and Journals. 13. Other International Journals. 14. Relevant conference proceedings and magazines (including ACM and IEEE conferences). 15. Technical reports from universities and major companies.
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